The ejection fraction (EF) is the portion of the end diastolic volume that is ejected from the ventricle during systole. It can be expressed as a decimal, but is more commonly multiplied by one hundred so that it can be expressed as a percentage. The normal ejection fraction is 55% of greater. The formula for ejection fraction is the stroke volume divided by the end diastolic volume. Because the stroke volume is calculated as the end diastolic volume minus the end systolic volume, the formula is usually expressed as

$$EF = \frac{\text{End Diastolic Volume} - \text{End Systolic Volume}}{\text{End Diastolic Volume}}$$

Calculate the ejection fraction for a patient with an end diastolic volume of 142 mL and an end systolic volume of 55 mL.

$$EF = \frac{EDV - ESV}{EDV} = \frac{142 - 55}{142} = \frac{87}{142} = 0.61 \text{ or } 61\%$$

Ejection Fraction Practice Problems
Calculate the ejection fraction for a patient with an

1. EDV of 131 and an ESV of 40
2. EDV of 105 and an ESV of 61
3. EDV of 124 and an ESV of 53
4. EDV of 144 and an ESV of 67
5. EDV of 96 and an ESV of 48
6. EDV of 115 and an ESV of 38
7. EDV of 119 and an ESV of 65
8. EDV of 130 and an ESV of 54
9. EDV of 100 and an ESV of 43
10. EDV of 186 and an ESV of 65

Ejection Fraction Calculation Answers

1. 69%
2. 42%
3. 57%
4. 53%
5. 50%
6. 67%
7. 45%
8. 58%
9. 57%
10. 65%

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